

ARISTO EDUCATIONAL PRESS  
HONG KONG DIPLOMA OF SECONDARY  
EDUCATION EXAMINATION

**(Mock Test 28)**

**Suggested Solution**

## Paper 1

|       |       |       |       |       |
|-------|-------|-------|-------|-------|
| 1. A  | 11. B | 21. C | 31. B | 41. B |
| 2. C  | 12. A | 22. B | 32. A | 42. C |
| 3. C  | 13. A | 23. A | 33. C | 43. A |
| 4. C  | 14. B | 24. B | 34. D | 44. D |
| 5. C  | 15. D | 25. C | 35. A | 45. B |
| 6. B  | 16. D | 26. B | 36. D |       |
| 7. A  | 17. D | 27. D | 37. C |       |
| 8. A  | 18. D | 28. B | 38. A |       |
| 9. C  | 19. C | 29. A | 39. A |       |
| 10. B | 20. B | 30. B | 40. C |       |

## Explanation

1. Answer: A

Suppose Kevin's order of preference remains unchanged. Kevin's opportunity cost of working as an intern at a TV station (the highest-valued option forgone) is the first option (to learn to drive a car).

Option A: This will reduce the value of the first option. The opportunity cost of working as an intern at a TV station will decrease.

Option B: This will increase the value of the first option. The opportunity cost of working as an intern at a TV station will increase.

Option C: This will reduce the value of the chosen option. The opportunity cost of working as an intern at a TV station will not be affected.

Option D: This is not the highest-valued option forgone, so the increase in its value will not reduce the opportunity cost of working as an intern at a TV station.

2. Answer: C

(2), (3): These statements make value judgements on the measures to prevent network attacks, and they are not refutable by facts, so they are normative statements.

(1): This statement describes the effect of network attacks with no value judgement, and it is refutable by facts, so it is a positive statement.

3. Answer: C

4. Answer: C

Since Ricky does not have to bear legal responsibility for the company, the form of business ownership of the company is a limited company.

Option C: Owners of a limited company enjoy limited liability.

Option A: If the company is a private limited company, it is not required to disclose its accounting information to the public.

Option B: If the company is a private limited company, it cannot issue shares and bonds to the public.

Option D: Since a limited company is a legal entity, it can still exist after the death of the owner.

5. Answer: C

These charging ports are excludable in consumption. When someone is using a charging port, it is possible to prevent others from using it, so it is a private good.

6. Answer: B

Since the percentage of employment in primary production decreased from 5% to 4%, it implies that the relative importance of primary production in terms of employment has decreased.

Options A and C: The question does not provide information on the number of employment in 2018 and 2021, so the changes in the number of employment and the number of workers engaged in the secondary production cannot be determined.

Option D: The question does not provide information about GDP.

7. Answer: A

Average labour productivity of firm A =  $100 \text{ units of output} / (15 \times 8) \text{ man-hours} = 0.83 \text{ units of output per man-hour}$

Average labour productivity of firm B =  $100 \text{ units of output} / (10 \times 8) \text{ man-hours} = 1.25 \text{ units of output per man-hour}$

As firm B's workers are more productive than that of firm A, we can infer that firm B probably provides more training for its workers.

Option B: Firm A's labour productivity is lower than that of firm B, showing that firm A's workers may not need to rush for more output to earn more income, so firm A may have adopted time rate.

Option C: If firm B uses less advanced machinery, its labour productivity will probably be lower, which is not consistent with the above calculation.

Option D: If firm A practises a higher degree of division of labour than firm B, its labour productivity will probably be higher, which is not consistent with the above calculation.

8. Answer: A

Occupational mobility refers to the ability and willingness of a factor of production to change from one occupation to another. Compared with the other three people, doctors need a lot of professional knowledge and they may have taken a long time to acquire the knowledge, so their opportunity cost of changing jobs will be higher. Therefore, they have the lowest occupational mobility.

9. Answer: C

(2): As the output of doctors is not standardised, the cost of measuring their output can be very high.

(3): The cost of monitoring treatment quality is lower because the doctors do not need to rush for more output.

(1): As doctors' incomes do not directly depend on output, paying them by time rate will not increase their productivity.

10. Answer: B

(3), (4): These are the fixed costs for the airlines, and they do not change with the number of flights.

(1), (2): These are the variable costs for the airlines, and they will decrease with fewer flights.

11. Answer: B

| Capital (units) | Labour (units) | Total product (units) | Marginal product (units) |
|-----------------|----------------|-----------------------|--------------------------|
| 5               | 1              | 10                    | 10                       |
| 5               | 2              | 21                    | 11                       |
| 5               | 3              | 33                    | 12                       |
| 5               | 4              | $Q_1$                 | X                        |
| 5               | 5              | $Q_2$                 | Y                        |

(2): When  $Q_1 = 49$  and  $Q_2 = 69$ ,  $X = 16$  and  $Y = 20$ . As marginal product keeps increasing, the above data does not illustrate the law of diminishing marginal returns.

(1): When  $Q_1 = 48$  and  $Q_2 = 60$ ,  $X = 15$  and  $Y = 12$ . When labour increases from 4 units to 5 units, marginal product decreases from 15 units to 12 units. The above data illustrates the law of diminishing marginal returns.

(3): When  $Q_1 = 40$  and  $Q_2 = 38$ ,  $X = 7$  and  $Y = -2$ . When labour increases from 3 units to 4 units, marginal product decreases from 12 units to 7 units. The above data illustrates the law of diminishing marginal returns.

12. Answer: A

More restaurants opening in the industrial building shows that the industry has expanded. A reduction in the marketing costs of the restaurants will cause the long run average cost to decrease, leading to external economies of scale.

13. Answer: A

| Output (units) | Original average cost (\$) | Original total cost (\$) | Original marginal cost (\$) | New average cost (\$) | New total cost (\$) | New marginal cost (\$) |
|----------------|----------------------------|--------------------------|-----------------------------|-----------------------|---------------------|------------------------|
| 1              | 20                         | 20                       | 15                          | 18                    | 18                  | 8                      |
| 2              | 21                         | 42                       | 22                          | 19                    | 38                  | 20                     |
| 3              | 22                         | 66                       | 24                          | 20                    | 60                  | 22                     |
| 4              | 23                         | 92                       | 26                          | 21                    | 84                  | 24                     |
| 5              | 24                         | 120                      | 28                          | 22                    | 110                 | 26                     |

Before the changes in the costs, when the market price is \$26, the profit-maximising output is 4 units.

Profit =  $\$26 \times 4 - \$92 = \$12$ .

After the changes in the costs, when the market price is \$26, the profit-maximising output is 5 units.

Profit =  $\$26 \times 5 - \$110 = \$20$ . Both the profit-maximising output and the profit will increase.

14. Answer: B

(1): When the government provides a per-unit subsidy on electric cars, the price of electric cars will decrease, and the quantity demanded will increase. If the demand for electric cars is elastic, the percentage decrease in price will be smaller than the percentage increase in quantity demanded. Total sales revenue will increase.

(3): When the government provides a per-unit subsidy on electric cars, the price of high-quality electric cars relative to low-quality electric cars will increase. A lower proportion of buyers will buy high-quality electric cars, so the ratio of high-quality electric cars sold to total electric cars sold will decrease.

(2): When the government provides a per-unit subsidy on electric cars, the price of electric cars will decrease. People will buy more electric cars, including high-quality electric cars.

15. Answer: D

(1), (2): These will increase the supply of labour, and the excess demand in the labour market will decrease.

(3): This will reduce the demand for labour, and the excess demand in the labour market will decrease.

16. Answer: D

More local workers retiring will cause the supply of local workers to decrease, shifting the supply curve to the left. The government increasing the quota for importing foreign workers will cause the wage rate of foreign workers to decrease. As local workers and foreign workers are substitutes, demand for local workers will decrease, shifting the demand curve to the left.

17. Answer: D

(1): Minibus service and other public transport services are substitutes. When the fee of other public transport services decreased, demand for minibus service decreased. There was an excess supply of minibus service.

(2): The quantity demanded of public transport services that offered free rides increased while the demand for them remained unchanged.

(3): The other public transport services are economic goods. When the price was zero, the quantity demanded would be larger than the quantity supplied, and there would be an excess demand.

18. Answer: D

How the tax burden is shared between consumers and producers depends on the relative magnitude of price elasticities of demand and supply. The more elastic party will have a smaller share of the tax burden.

Options A, B, C: Since the price elasticity of supply is higher than the price elasticity of demand, producers' tax burden will be smaller.

19. Answer: C

When the price ceiling is \$16, the price ceiling is ineffective. The price is \$14 and the quantity transacted is 20 units. Consumers' expenditure is  $\$14 \times 20 = \$280$ .

When the price ceiling is \$12, the price ceiling is effective. The quantity transacted will be equal to the quantity supplied, i.e. 15 units. Consumers' expenditure will be  $\$12 \times 15 = \$180$ .

Consumers' expenditure will decrease by \$100.

20. Answer: B

(1): Abolishing the effective quota on good X will cause its price to decrease while an increase in demand for good X will cause its price to increase. The change in the price is thus uncertain.

(3): Abolishing the effective quota on good X will eliminate the deadweight loss, the total social surplus will increase. An increase in demand for good X will cause the total social surplus to further increase.

(2): Consumer surplus will increase.

21. Answer: C

When the per-unit tax on good X decreases from  $t_2$  to  $t_1$ , the price will decrease from  $P_2$  to  $P_1$  and the quantity transacted will increase.

(2): The deadweight loss will decrease from area ach to area bcf. The deadweight loss will decrease by area abfh.

(1): The total tax revenue will change from area  $P_2ahg$  to area  $P_1bfe$ .

22. Answer: B

As offenders need to pay a fine, the fine increases the marginal private cost of smoking.

Option A: As the fine may not be equal to the marginal external cost, the quantity of cases of smoking may still be higher than the socially optimal level.

Option C: The fine is a government solution to the divergence between private and social costs.

Option D: Even if the fine reflects the external cost of smoking accurately, and the quantity of cases of smoking reduces to the socially optimal level which is lower than the equilibrium quantity, there may still be people smoking.

23. Answer: A

(1), (2): The high-income group will need to pay more tax while the low-income group will not be affected, the income gap will be narrowed. At the same time, aggregate demand will decrease, and the price level will decrease.

(3): Income of the low-income group will increase while the high-income group will not be affected, the income gap will be narrowed. However, aggregate demand will increase, and the price level will increase.

24. Answer: B

(1): The Gini coefficients based on original monthly household income was increasing, implying that the income gap between the rich and the poor was widening, and the income distribution became more uneven.

(3): The Gini coefficients based on post-tax post-social transfer monthly household income were lower than that based on original monthly household income, showing that Hong Kong's tax system and social transfers could reduce income inequality.

(2): The information provided cannot show the change in the amount of social transfers.

25. Answer: C

GDP at factor cost = Private consumption expenditure + Gross investment + Government consumption expenditure + Net exports – Indirect taxes + Subsidies

$$1,000 = 200 + (320 + 80) + X + 120 - 60 + 30$$

$$X = 310$$

26. Answer: B

As the percentage increase in per capita real GDP (3%) was higher than the percentage decrease in population (1%), real GDP has increased.

As real GDP has increased while nominal GDP has remained unchanged (the growth rate is 0%), the general price level has decreased.

27. Answer: D

A decrease in the price of raw materials will cause the short run aggregate supply to increase. A decrease in the salaries tax rate will cause private consumption expenditure to increase. Aggregate demand will increase.

Option A: The government increasing transfer payments will increase private consumption expenditure. Aggregate demand will increase. Increasing expenditure on education will increase government expenditure, and aggregate demand will increase. The economy's productive capacity will also increase. Both the short run aggregate supply and the long run aggregate supply will increase.

Option B: An increase in the number of tourists will cause net exports to increase. Aggregate demand will increase. An occurrence of a large-scale strike in the economy will cause the short run aggregate supply to decrease.

Option C: A decrease in asset demand for money will cause the interest rate to decrease. Investment and consumption will increase. Aggregate demand will increase. A decrease in the profits tax rate will cause investment expenditure to increase. Aggregate demand will increase.

28. Answer: B

Consumers expecting the future price level to increase will raise current consumption. Private consumption expenditure will increase. Aggregate demand will increase. The price level and aggregate output will increase in the short run.

Option A: An increase in people's desire to save will lead to a decrease in private consumption expenditure. Aggregate demand will decrease. The price level and aggregate output will decrease in the short run.

Option C: When the money supply decreases, the interest rate will increase. Consumption and investment will decrease. Aggregate demand will decrease. The price level and aggregate output will decrease in the short run.

Option D: When the economy's major trading partner experiences a recession, net exports will decrease. Aggregate demand will decrease. The price level and aggregate output will decrease in the short run.

29. Answer: A

When aggregate demand decreases, the long run equilibrium point may move from E to A.

Option A: When the government reduces the import tariff, imports will increase and net exports will decrease. There will be a deflationary gap. There will be an excess supply in the factor market, resulting in a downward pressure on input prices. A decrease in the cost of production will cause the short run aggregate supply to increase. In the long run, aggregate output will return to the full-employment output level with a lower price level.

Option B: The government distributing cash to citizens will increase private consumption expenditure. Aggregate demand will increase.

Option C: More residents emigrating overseas will cause aggregate demand, short run aggregate supply and long run aggregate supply to decrease. The long run equilibrium point will be on the new long run aggregate supply curve.

Option D: An increase in the cost of production will cause the short run aggregate supply to decrease.

30. Answer: B

(2): Restricted licence banks can only accept deposits of HK\$500,000 or above.

31. Answer: B

Withdrawing HK\$100,000 cash from a deposit-taking company will increase M1 and M2 by HK\$100,000, while M3 will remain unchanged.

Keeping half of the amount (\$50,000) as cash at home will not affect M1, M2 and M3.

Remitting the remainder (\$50,000) overseas will result in a decrease in M1, M2 and M3 by HK\$50,000, as this amount is no longer included in Hong Kong's money supply.

The overall effect of the above actions on M1, M2 and M3:

M1 will increase by HK\$50,000.

M2 will increase by HK\$50,000.

M3 will decrease by HK\$50,000.

32. Answer: A

Original reserves = Monetary base – Currency in public circulation = \$(100 – 20) million = \$80 million

Original deposits = Money supply – Currency in public circulation = \$(420 – 20) million = \$400 million

Original loans = \$(400 – 80) million = \$320 million

Required reserve ratio = (\$80 million/\$400 million)  $\times$  100% = 20%

After the public withdraw \$10 million from the banking system and hold it in cash:

New currency in public circulation = \$(20 + 10) million = \$30 million

New reserves = \$(80 – 10) million = \$70 million

Maximum amount of deposits = \$70 million  $\times$  (1/20%) = \$350 million

Option B: New money supply = \$(350 + 30) million = \$380 million. The money supply will decrease.

Option C: New monetary base = \$(30 + 70) million = \$100 million. The monetary base will remain unchanged.

Option D: Maximum amount of loans = \$(350 – 70) million = \$280 million. The maximum amount of loans will decrease by \$40 million.

33. Answer: C

A decrease in the discount rate will increase the money supply, the nominal interest rate will decrease and the quantity of money will increase.

When the liquidity of interest-bearing assets increases, the money demand will decrease. The nominal interest rate will decrease, and the quantity of money will remain unchanged.

In conclusion, the nominal interest rate will decrease, and the quantity of money will increase.

34. Answer: D

Expected real interest rate (Expected real rate of return of the bond) = Nominal interest rate – Expected inflation rate

Actual real interest rate (Actual real rate of return of the bond) = Nominal interest rate – Actual inflation rate

When the actual inflation rate is higher than expected, with the nominal interest rate remaining unchanged, the actual real interest rate will be lower than the expected real interest rate.

35. Answer: A

Quantity theory of money:  $MV = PY$ , where  $M$  = money supply,  $V$  = velocity of circulation of money,  $P$  = price level and  $Y$  = real output, assuming  $V$  is constant.

$\% \text{ change in } M \approx \% \text{ change in } P + \% \text{ change in } Y \approx \% \text{ change in nominal output}$

(1): As the growth rate of  $M$  is increasing, the growth rate of nominal output is also increasing, showing that nominal output is increasing.

(2): When the growth rate of  $M$  is increasing and higher than that of  $P$ , the growth rate of  $Y$  will be positive, showing that the real output is increasing.

36. Answer: D

More firms expanding their businesses implies that the number of job positions will increase. The unemployed population will decrease. The unemployment rate will decrease.

Option A: If the government relaxes the policy on importing foreign workers, the labour force will increase. As the imported foreign workers may replace local workers, the unemployed population may increase. The unemployment rate may increase.

Option B: The increase in unemployment benefits will reduce people's incentive to work. The unemployed population will increase and the unemployment rate will increase.

Option C: The reduction in the number of places in post-secondary education will cause more young people to enter the labour market as they cannot continue their studies. The labour force will increase. However, some of them may not be able to find a job, so the change in the unemployment rate is uncertain

37. Answer: C

Hong Kong adopts a source principle of taxation. Only income and profits earned in Hong Kong are subject to tax.

(1): Since Ms. Cheung's rental income is earned in the Mainland, it is not subject to tax in Hong Kong.

38. Answer: A

Under the new tax structure, the tax rate paid by the high-income group will increase. This will reduce their work incentive.

Option B: As the tax allowance remains unchanged, the tax base does not change.

Option C: Under the new tax structure, the tax rate paid by the high-income group increases while the tax rate paid by the low-income group decreases, thus narrowing the income gap between the rich and the poor.

Option D: The change in government tax revenue is uncertain.

39. Answer: A

(2): Sales tax is a regressive tax. The average tax rate paid by the high-income group is lower than that of the low-income group. Imposing a sales tax will lead to a more uneven distribution of income.

(1): Imposing a sales tax will cause people's consumption to decrease. GDP will decrease.

(3): The effect of imposing a sales tax on government tax revenue is uncertain, and the information about government revenue and expenditure is not given, so we cannot determine if the government will have a fiscal surplus.

40. Answer: C

As the short run equilibrium output level falls below the full-employment output level, there is a deflationary (output) gap.

Lowering the profits tax rate will increase investment expenditure, causing the aggregate demand to increase. The aggregate demand curve will shift to the right.

Option D: An open market sale of government bonds will cause the money supply to decrease. Interest rate will increase. Consumption and investment will decrease. Aggregate demand will decrease. The short run equilibrium output level will be farther away from the full-employment output level.

41. Answer: B

(2): A decrease in the basic allowance of the salaries tax may increase government revenue, improving the fiscal deficit. It will reduce people's disposable income, causing imports to decrease, improving trade deficit.

(1): An increase in the money supply will cause the interest rate to decrease. Consumption and investment will increase. Aggregate demand will increase. An increase in national income will cause government tax revenue to increase, improving the fiscal deficit. However, an increase in national income will lead to an increase in imports, worsening the trade deficit.

(3): An appreciation of Country A's currency will cause its export price (in terms of foreign currency) to increase, the volume of exports will decrease. The export value (in terms of Country A's currency) will decrease. At the same time, the import price (in terms of Country A's currency) will decrease and the volume of imports will increase. The change in the import value (in terms of Country A's currency) will be uncertain, depending on the price elasticity of demand. The effect on trade deficit will thus be uncertain. The decrease in export value will negatively affect the business of export-related industries, decreasing government profits tax revenue and worsening the fiscal deficit.

42. Answer: C

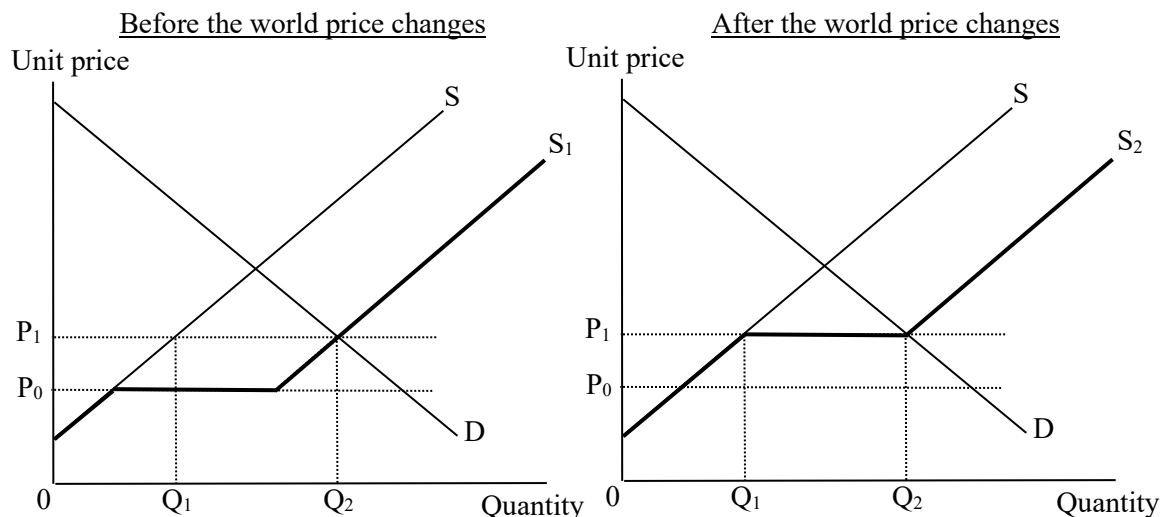
Country A's opportunity cost of producing 1 unit of milk =  $(40 / 20)$  units of juice = 2 units of juice

Country B's opportunity cost of producing 1 unit of milk =  $(16 / 20)$  units of juice = 0.8 units of juice

For the terms of trade of milk to be mutually beneficial, its range should be

0.8 units of juice < 1 unit of milk < 2 units of juice.

43. Answer: A



$S_1$  is the supply curve before the world price changes while  $S_2$  is the supply curve after the world price changes.

Option A: The domestic price will remain unchanged at  $P_1$ .

Option B: The import volume will remain unchanged at  $(Q_2 - Q_1)$  units.

Option C: As the domestic price does not change, domestic production will remain unchanged at  $Q_1$ .

Option D: As the quantity supplied equals the quantity demanded, there is no shortage of the good before and after the world price changes.

44. Answer: D

(2): Invisible trade balance =  $\$(15 - 25)$  million =  $-\$10$  million

(3): Current account balance = Visible trade balance + Invisible trade balance + Net factor income from abroad + Net current transfers

$$= \$[(50 - 10) + (-10) + (40 - 20) + (10 - 25)] \text{ million} \\ = \$35 \text{ million}$$

(4): As there is a decrease in reserve assets of  $\$20$  million, the balance of payments deficit is  $\$20$  million.

(1): As the income receivable from abroad is larger than the income payable abroad, the net factor income from abroad is positive. The economy's gross national income is larger than its gross domestic product.

45. Answer: B

(1): To issue Hong Kong dollar banknotes, the note-issuing banks have to submit an equivalent amount of US dollars in exchange for the Certificate of Indebtedness issued by the Exchange Fund at the linked rate. Therefore, Hong Kong's monetary base (i.e. cash in public circulation + bank reserves) is fully backed by US dollars.

(4): Under the linked exchange rate system, when the US dollar depreciates against other currencies, the Hong Kong dollar will depreciate against other currencies accordingly. The price of Hong Kong exports (in terms of other currencies) will decrease, the volume of exports will increase. The total value of Hong Kong exports (in terms of Hong Kong dollars) will increase.

(2): Only the official exchange rate between the Hong Kong Monetary Authority and the note-issuing banks is fixed at HK\$1 to HK\$7.8.

(3): Under the linked exchange rate system, when the US dollar depreciates against other currencies, the Hong Kong dollar will depreciate against other currencies accordingly. The price of Hong Kong imports (in terms of Hong Kong dollars) will increase. The average price of Hong Kong imports will tend to increase.

## Paper 2

### Section A (44 marks)

1. (a) Advantages:

- The cost of supervision is lower because workers tend to work harder to earn more income.
- Part of the business risk is shared by the workers.
- Any reasonable answers

2@, max: 4 marks

(Mark the **FIRST TWO** points only)

- (b) Law of diminishing marginal returns. The law states that in the short run, as more units of a variable factor are added to a given quantity of fixed factor successively, marginal product will decrease eventually, other things being constant. In this case, the workers are variable factors while the trucks are fixed factors. (5 marks)

2. The law of demand states that an increase (a decrease) in the price of a good will result in a decrease (an increase) in its quantity demanded, *ceteris paribus*.

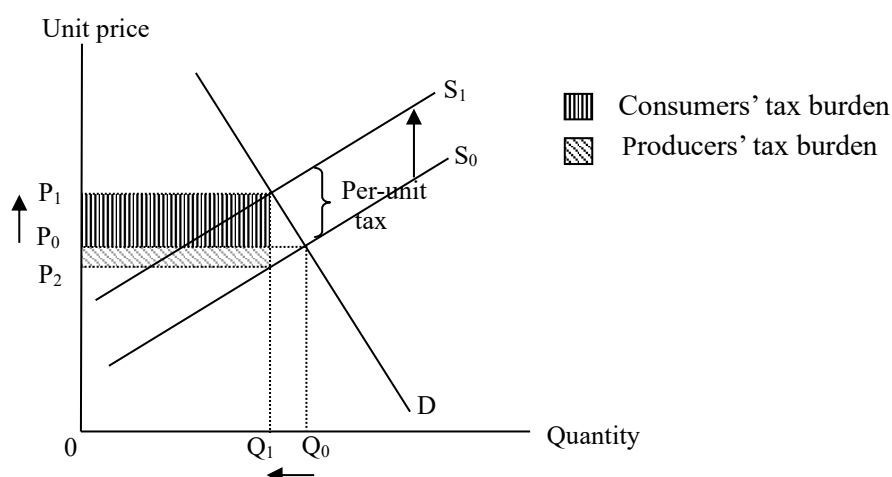
After the price of a snack is added, the relative price of a full day pass would decrease in terms of a half day pass. According to the law of demand, people would buy more full day passes relative to half day passes, so the ratio of full day passes sold to half day passes sold would increase. (5 marks)

3. Verbal elaboration:

If the demand is less elastic than the supply, the consumers' tax burden will be greater than the producers' tax burden. (1 mark)

Illustrate in the diagram:

- upward shift of the supply curve (1 mark)
- price increases and quantity transacted decreases (1 mark)
- correct positions of the consumers' tax burden and producers' tax burden (1 mark)
- consumers' tax burden > producers' tax burden (1 mark)



4. (a) Yes. (1 mark)  
The music is non-rival in consumption. When someone listens to the music on a music streaming platform, the amount of music available to others will not be reduced. (2 marks)
- (b) No. (1 mark)  
The service is produced from scarce resources which have alternative uses. / More of the service is preferred. (1 mark)
5. (a) Required reserve ratio =  $[(\$3,000 - \$1,000) / 8,000] \times 100\% = 25\%$  (1 mark)
- (b) (i) No, because the decrease in reserves is offset by the increase in cash in public circulation. (2 marks)
- (ii) Maximum possible amount of money supply =  $\$1,500 + [(\$3,000 - \$1,500) \times 1/25\%]$   
= \$7,500 (3 marks)
6. (a) The opportunity cost of producing 1 unit of juice in Country A = 3 (= 27/9) units of bread (1 mark)  
The opportunity cost of producing 1 unit of juice in Country B = 2.5 (= 15/6) units of bread (1 mark)  
Country B has a comparative advantage in producing juice as it has a lower opportunity cost in producing juice. (1 mark)
- (b) Total output of bread = 10 (= 90/9) units of bread. (1 mark)  
Total output of juice = 6 (= 90/15) units of juice. (1 mark)
- (c) Terms of trade: 1 unit of juice = 2.8 (= 14/5) units of bread (1 mark)  
Gain from trade of Country B per unit of juice exported =  $(2.8 - 2.5)$  units of bread = 0.3 units of bread (1 mark)  
Total gain from trade from exporting 5 units of juice =  $0.3 \text{ units of bread} \times 5 = 1.5 \text{ units of bread}$  (1 mark)
7. (a) (Visible) trade balance in 2018 = -HK\$252,997 million (= HK\$4,453,350 million - HK\$4,706,347million)  
(Visible) trade balance in 2019 = -HK\$120,521 million (= HK\$4,255,098 million - HK\$4,375,619 million)  
During the period, (visible) trade deficit decreased. (2 marks)
- (b) When the US dollar depreciated against a basket of currencies, the Hong Kong dollar depreciated against a basket of currencies accordingly under the linked exchange rate system. The price of imported goods (in terms of Hong Kong dollars) would increase and Hong Kong would import fewer foreign goods. As the demand for imported goods is elastic, the total value of imports of goods (in terms of Hong Kong dollars) would decrease. (4 marks)

## Section B (60 marks)

8. (a) Reasons:

- Masks from different mask factories are heterogeneous.
- Mask factories are price searchers. Each factory sets its own price.
- Market information is imperfect, e.g. information about the quality of masks.
- Non-price competition exists, e.g. advertisements.
- Any reasonable answers

2@, max: 2 marks

(Mark the **FIRST** point only)

(b) Verbal elaboration:

As the production cost decreases, the supply of masks will increase. (1 mark)

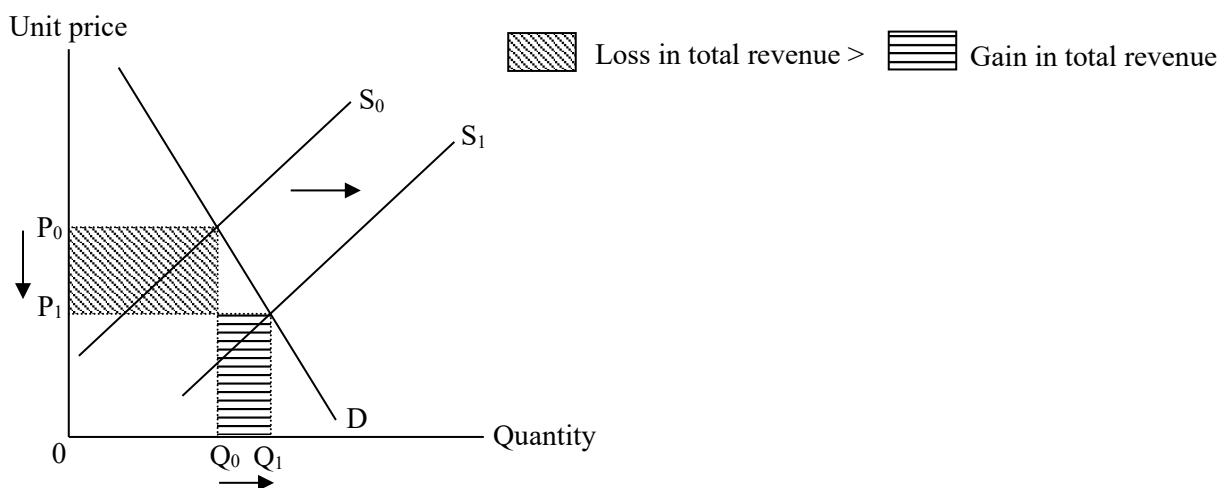
If the demand for masks is inelastic, the percentage decrease in price will be greater than the percentage increase in quantity demanded. Total revenue will decrease. (3 marks)

Illustrate in the diagram:

- rightward shift of the supply curve (1 mark)

- correct positions of gain in total revenue and loss in total revenue (1 mark)

- gain in total revenue < loss in total revenue (1 mark)



(c) A reason:

- Not all factor inputs used to produce masks came from Hong Kong, e.g. some raw materials were imported.
- Some of the masks were not produced in the current year, e.g. some of them came from inventory.
- Any reasonable answer

2@, max: 2 marks

(Mark the **FIRST** point only)

(d) Wearing a mask can reduce the chance of others being infected with viruses and bring benefits to others, but those who wear a mask do not receive any compensation. There is an external benefit. The behaviour of wearing a mask may involve positive externality. (3 marks)

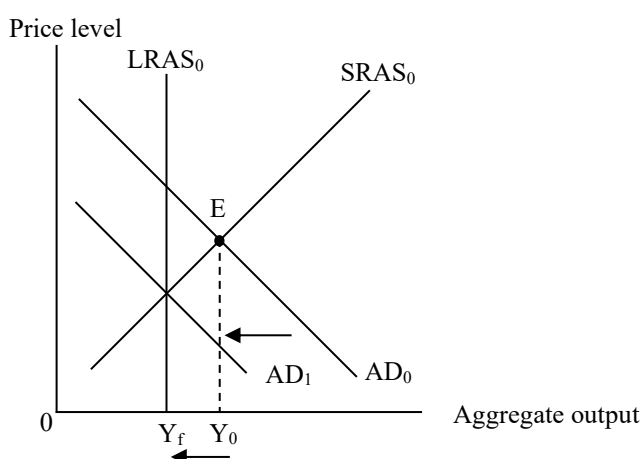
9. (a) There is an inflationary (output) gap as the aggregate output is above the full-employment output level. (2 marks)

(b) Verbal elaboration:

- A contractionary monetary policy will lead to a decrease in the money supply. (1 mark)
- The interest rate will increase. (1 mark)
- Consumption/investment will decrease. (1 mark)
- Aggregate demand will decrease. (1 mark)
- Aggregate output will decrease and restore to the full-employment output level. (1 mark)

Illustrate in the diagram:

- leftward shift of the AD curve (1 mark)
- aggregate output decreases to the full-employment output level ( $Y_f$ ) (1 mark)



(c) An advantage:

- Raising the salaries tax rate may increase government tax revenue. The fiscal balance may improve. Imposing a contractionary monetary policy will not affect the fiscal balance of the government.

2@, max: 2 marks

- Any reasonable answer

(Mark the **FIRST** point only)

A disadvantage:

- When the salaries tax rate increases, the net income of workers decreases. It may reduce the incentive to work and lead to a fall in the labour supply. Imposing a contractionary monetary policy will not affect people's incentive to work.

2@, max: 2 marks

- Any reasonable answer

(Mark the **FIRST** point only)

(d) A policy:

- Raising the minimum wage rate: The production cost of firms will increase, and the short run aggregate supply will decrease.

2@, max: 2 marks

- Any reasonable answer

(Mark the **FIRST** point only)

10. (a) Capital. (1 mark)

(b) Lateral expansion. (1 mark)

Ocean Park has expanded its business into the production of related but not directly competing goods. (1 mark)

(c) (i) No. The full cost of visiting the Water World includes the admission fare (money cost) and the time cost. On weekdays, although the admission fare of the Water World is lower than that on weekends, the time cost may be higher. If the extra time cost is higher than the money cost saved, the full cost of visiting the Water World on weekdays will be higher. (4 marks)

(ii) Rationing function. (1 mark)

Tickets on public holidays are distributed to buyers who are willing to pay a higher price. (1 mark)

(d) Verbal elaboration:

Both the demand and supply of the tickets of Ocean Park decreased. (2 marks)

The price remained unchanged and the quantity transacted decreased. (1 mark)

Admission income decreased. (1 mark)

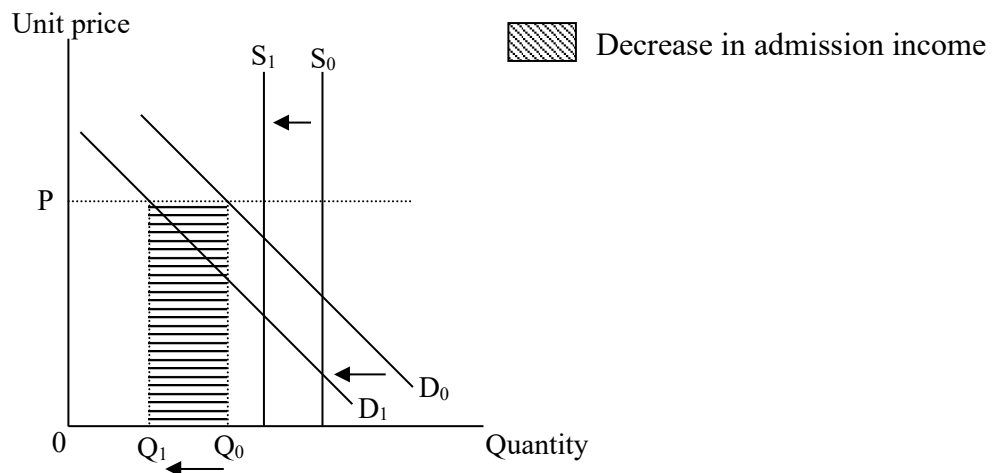
Illustrate in the diagram:

- leftward shift of the demand curve (1 mark)

- leftward shift of the supply curve (1 mark)

- price remains unchanged, quantity transacted decreases (1 mark)

- admission income decreases (1 mark)



(e) Marks award for effective communication (EC: max 2 marks)

| Marks | Performance                                                                                                                                                                                                                                                                                                                                                     |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2     | <ul style="list-style-type: none"><li>● Arguments supported with the source/data and appropriate economic theories</li><li>● Relevant materials presented</li><li>● Well-organised and coherent answers presented without repetition of ideas</li><li>● Ideas clearly and fluently expressed with appropriate use of language/words/terms/symbols</li></ul>     |
| 1     | <ul style="list-style-type: none"><li>● Arguments presented with some support of the source/data and economic theories</li><li>● Some irrelevant materials presented</li><li>● Answers presented in a less organised way with some repetition of ideas</li><li>● A clear message conveyed with some inappropriate use of language/words/terms/symbols</li></ul> |
| 0     | <ul style="list-style-type: none"><li>● Arguments presented with no support of the source/data and economic theories</li><li>● Materials unrelated to the gist of the question presented</li><li>● Inconsistent arguments presented</li><li>● Limited ideas expressed with inappropriate use of words/terms/symbols</li></ul>                                   |

The maximum marks for content are 12 marks. Answers may include the following:

Effects on aggregate output:

- If the park closes down, the number of tourists visiting Hong Kong may decrease. Exports of services will decrease and net exports will decrease. Aggregate demand will decrease. In the short run, aggregate output will decrease.
- If the park closes down, Hong Kong's productive capacity will decrease. Long run aggregate supply will decrease. In the long run, the full-employment output level will decrease.

Effects on fiscal balance:

- The closing down of Ocean Park may reduce the number of tourists visiting Hong Kong, the profits of hotels and restaurants will decrease. The profits tax revenue of the Hong Kong government will decrease, which will worsen the fiscal balance.
- If the park closes down, the government does not need to provide relief funds to Ocean Park any more. The financial burden of the government will not increase.

Effects on unemployment rate:

- Ocean Park Corporation employs many workers. If Ocean Park closes down, the workers may face unemployment. The unemployed population will increase. Assume the labour force remains unchanged, the unemployment rate in Hong Kong will increase.

Effects on other economic variables:

- Hong Kong's gross domestic product may decrease. If Ocean Park closes down, private consumption expenditure may decrease. The number of tourists may decrease, causing the exports of services to decrease. Net exports will decrease.
- Hong Kong's balance of payments may worsen. If Ocean Park closes down, the number of tourists may decrease, and the exports of services will decrease.

**Section C (16 marks)**

11. (a) Illustrate in the diagram:

- correct position of MR (1 mark)
- correct position of  $P_M$  (1 mark)
- correct position of  $Q_M$  (1 mark)

(b) Verbal elaboration:

At the profit-maximising output level ( $Q_M$ ), the marginal benefit is greater than the marginal cost.

(1 mark)

Illustrate in the diagram:

- correct position of deadweight loss (1 mark)

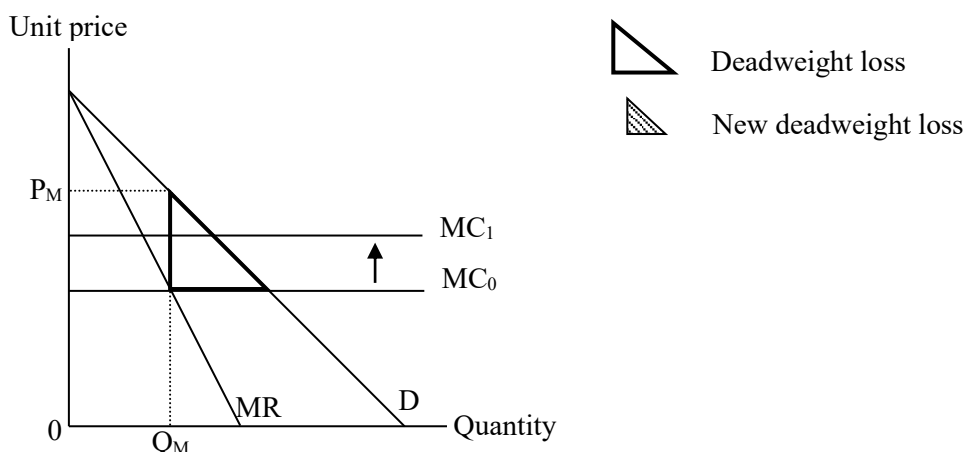
(c) Verbal elaboration:

Economic efficiency will be improved and the deadweight loss will decrease, as the gap between marginal benefit and marginal cost will be narrowed and the output level will be closer to the efficient level.

(2 marks)

Illustrate in the diagram:

- correct position of the new deadweight loss/correct position of the decrease in deadweight loss/decrease in the gap between marginal benefit and marginal cost (1 mark)



(d) No, all consumers are facing the same pricing arrangement and there is no market segmentation.

(3 marks)

(e) Market sharing / market allocation.

(1 mark)

Motives:

- It would face less competition in Country A.
- It can raise prices and earn a higher profit in Country A.
- Any reasonable answers

} 1 @, max: 2 marks

(Mark the **FIRST TWO** points only)

(f) A reason:

- Lok's Amusement Park is a natural monopoly and its output is enough to satisfy market demand.
- Any reasonable answer

} 2 @, max: 2 marks

(Mark the **FIRST** point only)

12. (a) The marginal costs of producing both good X and good Y are increasing in Country A. (2 marks)

(b) Verbal elaboration:

- The slope of L is the opportunity cost of producing good X.

(1 mark)

- AB is steeper than L.

(1 mark)

- This implies that the opportunity cost of producing good X is lower than the world price.

(1 mark)

- Therefore, Country A has a comparative advantage in producing good X.

(1 mark)

max: 3 marks

Illustrate in the diagram:

- correct position of the straight line L

(1 mark)

(c) (i) Illustrate in the diagram:

- correct position of  $P_T$

(1 mark)

- tangency between  $CPF_A$  and  $PPF_A$

(1 mark)

-  $CPF_A$  is parallel to AB

(1 mark)

(ii) Verbal elaboration:

Country A can consume beyond its production possibility frontier.

(1 mark)

Illustrate in the diagram:

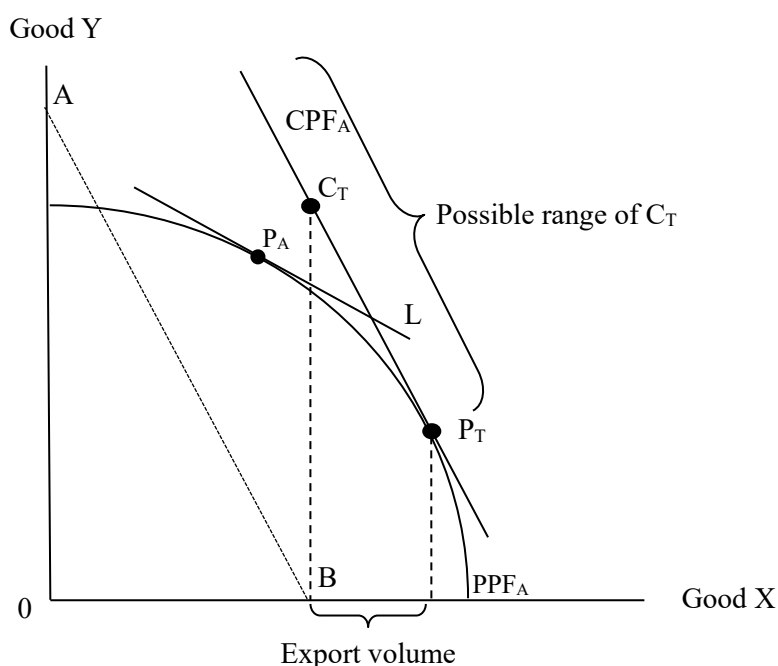
- correct position of  $C_T$

(1 mark)

(iii) Illustrate in the diagram:

- correct position of export volume

(1 mark)



(d) (i) A reason:

- Productivity may increase due to the import of superior technology from trading partners, leading to a higher GNI per capita.

- Any reasonable answer

(Mark the **FIRST** point only)

} 2@, max: 2 marks

(ii) A reason:

- Pollution has become more serious.

- Income inequality has become more serious.

- Any reasonable answers

(Mark the **FIRST TWO** points only)

} 1@, max: 2 marks